

Aditya Kusupati

Google DeepMind
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RESEARCH INTERESTS

I focus on designing fundamental *Machine Learning* algorithms with strong empirical performance & real-world deployability geared towards enabling **adaptive intelligence**. They have been widely adopted by industry serving over a Billion users daily and open-source communities. My research is used extensively across Google – Gemini (2.0 for AI Overviews, 1.5 Flash/Flash-8B/Nano), Gemma 3n, Gemini Embeddings, Search Representations, Ads Models and GenMedia Models. It has also been widely deployed across industry (most dense embeddings) and open source (~10% Hugging Face downloads).

EDUCATION

University of Washington, Seattle 2019 - 2024
PhD in Computer Science and Engineering
Visiting Research Associate in Harvard Computer Science 2023 - 2024
Advisors: Prof. Ali Farhadi & Prof. Sham Kakade
Committee: Prof. Luke Zettlemoyer, Prof. Zaid Harchaoui & Dr. Rahul Sukthankar

Indian Institute of Technology Bombay 2013 - 2017
B.Tech (Honours) in Computer Science and Engineering with Minor in Electrical Engineering
Advisor: Prof. Soumen Chakrabarti

WORK EXPERIENCE

Google DeepMind July 2024 - Present
Staff Research Scientist
Manager: Dr. Rahul Sukthankar
Leading multiple long-term research efforts with a team of interns, pre-doc researchers, research engineers and scientists.

Google Research → DeepMind January 2022 - June 2024
Student Researcher in Perception
Advisors: Dr. Prateek Jain & Tom Duerig
Worked on making fundamental Machine Learning algorithms to make AI models and systems accurate and performant.

NVIDIA Toronto AI Lab June - September, 2020
Research Scientist Intern
Advisor: Prof. Sanja Fidler
Explored missing modality (audio/visual/control) generation with cross-modal supervision for Atari video games using GANs.

Microsoft Research India May 2017 - July 2019
Research Fellow in Machine Learning and Optimization
Advisors: Dr. Manik Varma & Dr. Prateek Jain
Worked on resource-efficient and large-scale machine learning resulting in top-tier publications & deployment in Bing.

PUBLICATIONS

Preprints

* - equal contribution

2. **Compressing Many-Shots in In-Context Learning**

Devvrit, Pranamya Prashant Kulkarni, Nilesh Gupta, Yerram Varun, Liqian Peng, Jay Yagnik, Praneeth Netrapalli, Cho-Jui Hsieh, Alec Go, Inderjit S Dhillon, **Aditya Kusupati**, Prateek Jain.
Under Review, 2025.

1. **MatMamba: A Matryoshka State Space Model**

Abhinav Shukla, Sai Vemprala, **Aditya Kusupati**, Ashish Kapoor.
Under Review, 2025.

Conference Publications

25. **CuRe: Cultural Gaps in the Long Tail of Text-to-Image Models**

Aniket Rege, Zinnia Nie, Unmesh Raskar, Mahesh Ramesh, Zhuoran Yu, **Aditya Kusupati***, Yong Jae Lee*, Ramya Korlakai Vinayak*.
International Conference on Computer Vision ICCV, 2025
Workshop on Demographic Diversity in Computer Vision @ CVPR 2025 (Oral).

24. **Matryoshka Quantization**
Pranav Nair*, Puranjay Datta*, Jeff Dean, Prateek Jain **Aditya Kusupati**.
International Conference on Machine Learning (ICML), 2025.
Workshop on Sparsity in LLMs @ ICLR 2025 (Oral).
23. **ActionAtlas: A VideoQA Benchmark for Fine-grained Action Recognition**
Mohammadreza Salehi, Jae Sung Park, Tanush Yadav, **Aditya Kusupati**,
Ranjay Krishna, Yejin Choi, Hannaneh Hajishirzi, Ali Farhadi.
Neural Information Processing Systems (NeurIPS) Dataset and Benchmarks Track, 2024.
22. **Superposed Decoding: Multiple Generations from a Single Autoregressive Inference Pass**
Ethan Shen, Alan Fan, Sarah Pratt, Jae Sung Park, Matthew Wallingford, Sham Kakade, Ari Holtzman,
Ranjay Krishna, Ali Farhadi, **Aditya Kusupati**.
Neural Information Processing Systems (NeurIPS), 2024.
21. **From an Image to a Scene: Learning to Imagine the World from a Million 360° Videos**
Matthew Wallingford, Anand Bhattad, **Aditya Kusupati**, Vivek Ramanujan, Matt Deitke, Sham Kakade, Aniruddha
Kembhavi, Roozbeh Mottaghi, Wei-Chiu Ma, Ali Farhadi.
Neural Information Processing Systems (NeurIPS), 2024.
20. **Mixture of Nested Experts: Adaptive Processing of Visual Tokens**
Gagan Jain, Nidhi Hegde, **Aditya Kusupati**, Arsha Nagrani, Shyamal Buch,
Prateek Jain, Anurag Arnab, Sujoy Paul.
Neural Information Processing Systems (NeurIPS), 2024.
19. **MatFormer: Nested Transformer for Elastic Inference**
Devvrit*, Sneha Kudugunta*, **Aditya Kusupati***, Tim Dettmers, Kaifeng Chen,
Inderjit Dhillon, Yulia Tsvetkov, Hannaneh Hajishirzi, Sham Kakade, Ali Farhadi and Prateek Jain.
Neural Information Processing Systems (NeurIPS), 2024.
Efficient Natural Language and Speech Processing workshop @ NeurIPS 2023 (Oral, 🏆 Best Paper Award).
Workshop on Advancing Neural Network Training @ NeurIPS 2023 (Oral).
18. **EHF: End-to-end learning of Hierarchical Index for Efficient Dense Retrieval**
Ramnath Kumar*, Anshul Mittal*, Nilesh Gupta, **Aditya Kusupati**, Inderjit Dhillon, Prateek Jain.
Transactions on Machine Learning Research (TMLR), 2024.
17. **Gecko: Versatile Text Embeddings Distilled from Large Language Models**
Jinhyuk Lee*, Zhuyun Dai*, Xiaoqi Ren*, Blair Chen, Daniel Cer, Jeremy R. Cole, Kai Hui, Michael Boratko, Rajvi
Kapadia, Wen Ding, Yi Luan, Sai Meher Karthik Duddu, Gustavo Hernandez Abrego, Weiqiang Shi, Nithi Gupta,
Aditya Kusupati, Prateek Jain, Siddhartha Reddy Jonnalagadda, Ming-Wei Chang, Iftexhar Naim.
Google Technical Report, 2024.
16. **SHARCS: Efficient Transformers through Routing with Dynamic Width Sub-networks**
Mohammadreza Salehi, Sachin Mehta, **Aditya Kusupati**, Ali Farhadi, Hanna Hajishirzi.
Empirical Methods in Natural Language Processing (EMNLP) Findings, 2023.
15. **Objaverse-XL: A Universe of 10M+ 3D Objects**
Matt Deitke, Ruoshi Liu, Matthew Wallingford, Huong Ngo, Oscar Michel, **Aditya Kusupati**, Alan Fan, Christian
Laforte, Vikram Voleti, Samir Yitzhak Gadre, Aniruddha Kembhavi, Carl Vondrick, Georgia Gkioxari, Kiana Ehsani,
Ludwig Schmidt, Ali Farhadi.
Neural Information Processing Systems (NeurIPS) Dataset and Benchmarks Track, 2023.
14. **MADLAD-400: Monolingual And Document-Level Large Audited Dataset**
Sneha Kudugunta, Isaac Caswell, Biao Zhang, Xavier Garcia, Christopher A. Choquette-Choo, Katherine Lee, Derrick
Xin, **Aditya Kusupati**, Romi Stella, Ankur Bapna, Orhan Firat.
Neural Information Processing Systems (NeurIPS) Dataset and Benchmarks Track, 2023.
13. **Neural Priming for Sample-Efficient Adaptation**
Matthew Wallingford*, Vivek Ramanujan*, Alex Fang, **Aditya Kusupati**,
Roozbeh Mottaghi, Aniruddha Kembhavi, Ludwig Schmidt, Ali Farhadi.
Neural Information Processing Systems (NeurIPS), 2023.
12. **AdANNS: A Framework for Adaptive Semantic Search**
Aniket Rege*, **Aditya Kusupati***, Sharan Ranjit, Alan Fan, Qingqing Cao,
Sham Kakade, Prateek Jain, Ali Farhadi.
Neural Information Processing Systems (NeurIPS), 2023.
Practical ML for Developing Countries workshop @ ICLR 2023 (Oral).

11. **FLUID: A Unified Evaluation Framework for Flexible Sequential Data**
Matthew Wallingford, **Aditya Kusupati***, Keivan Alizadeh-Vahid*, Aaron Walsman, Aniruddha Kembhavi, Ali Farhadi.
Transactions on Machine Learning Research (TMLR), 2023.
10. **Neural Radiance Field Codebooks**
Matthew Wallingford, **Aditya Kusupati**, Alex Fang, Vivek Ramanujan, Aniruddha Kembhavi, Roozbeh Mottaghi, Ali Farhadi
International Conference on Learning Representations (ICLR), 2023.
9. **Matryoshka Representation Learning.**
Aditya Kusupati*, Gantavya Bhatt*, Aniket Rege*, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain,, Ali Farhadi.
Neural Information Processing Systems (NeurIPS), 2022.
Vision Transformers: Theory and Applications workshop @ NeurIPS, 2022 (Oral).
Self-Supervised Learning - Theory and Practice workshop @ NeurIPS, 2022.
Computer Vision in the Wild workshop @ ECCV, 2022.
8. **MERLOT RESERVE: Neural Script Knowledge through Vision and Language and Sound**
Rowan Zellers, Jiasen Lu, Ximing Lu, Youngjae Yu, Yanpeng Zhao, Mohammadreza Salehi, **Aditya Kusupati**, Jack Hessel, Ali Farhadi, Yejin Choi.
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022 (Oral).
7. **ProtoSound: Personalized, Scalable Sound Recognition for d/Deaf and Hard of Hearing Users through In-the-Wild Few-Shot Interactions.**
Dhruv Jain, Khoa Nguyen, Steven Goodman, Rachel Grossman-Kahn, Hung Ngo, **Aditya Kusupati**, Ruofei Du, Alex Olwal, Leah Findlater, Jon Froehlich.
ACM CHI Conference on Human Factors in Computing Systems (CHI), 2022 (Talk).
6. **LLC: Accurate, Multi-purpose Learnt Low-dimensional Binary Codes**
Aditya Kusupati, Matthew Wallingford, Vivek Ramanujan, Raghav Somani, Jae Sung Park, Krishna Pillutla, Prateek Jain, Sham Kakade, Ali Farhadi.
Neural Information Processing Systems (NeurIPS), 2021 (Virtual Talk).
5. **RNNPool: Efficient Non-linear Pooling for RAM Constrained Inference**
Oindrila Saha, **Aditya Kusupati**, Harsha Vardhan Simhadri, Manik Varma, Prateek Jain.
Neural Information Processing Systems (NeurIPS), 2020 (Virtual Spotlight).
WiCV workshop @ CVPR, 2020.
4. **Soft Threshold Weight Reparameterization for Learnable Sparsity**
Aditya Kusupati, Vivek Ramanujan*, Raghav Somani*, Mitchell Wortsman*, Prateek Jain, Sham Kakade, Ali Farhadi.
International Conference on Machine Learning (ICML), 2020 (Virtual Talk).
3. **Extreme Regression for Dynamic Search Advertising**
Yashoteja Prabhu, **Aditya Kusupati**, Nilesch Gupta, Manik Varma.
International Conference on Web Search and Data Mining (WSDM), 2020 (Long Oral).
eXtreme Classification: Theory and Applications workshop @ ICML, 2020.
2. **One Size Does Not Fit All: Multi-Scale, Cascaded RNNs for Radar Classification**
Dhrubojyoti Roy*, Sangeeta Srivatsava*, **Aditya Kusupati**, Pranshu Jain, Manik Varma, Anish Arora.
International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2019.
🏆 Best Paper Runner-Up Award.
1. **FastGRNN: A Fast, Accurate, Stable and Tiny Kilobyte Sized Gated Recurrent Neural Network**
Aditya Kusupati, Manish Singh, Kush Bhatia, Ashish Kumar, Prateek Jain, Manik Varma.
Neural Information Processing Systems (NeurIPS), 2018.

Workshop Publications

2. **Are “Hierarchical” Visual Representations Hierarchical?**
Ethan Shen, Ali Farhadi, **Aditya Kusupati**.
Workshop on Symmetry and Geometry in Neural Representations @ NeurIPS 2023.
1. **Disrupting Model Training with Adversarial Shortcuts**
Ivan Evtimov, Ian Covert, **Aditya Kusupati**, Tadayoshi Kohno.
Workshop on Adversarial Machine Learning @ ICML 2021.

Journal Publications

1. **One Size Does Not Fit All: Multi-Scale, Cascaded RNNs for Radar Classification**
Dhrubojyoti Roy*, Sangeeta Srivatsava*, **Aditya Kusupati**, Pranshu Jain, Manik Varma, Anish Arora.
ACM Transactions on Sensor Networks (TOSN), 17(2), January 2021. (**Best Paper Nomination**).

Demos

1. **Lightweight, Deep RNNs for Radar Classification**
Dhrubojyoti Roy*, Sangeeta Srivatsava*, Pranshu Jain, **Aditya Kusupati**, Manik Varma, Anish Arora.
International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2019.

Theses

2. **Towards Adaptive Intelligence**
Aditya Kusupati.
PhD Thesis, Paul G. Allen School of Computer Science and Engineering, University of Washington, 2019 - 24.
1. **Efficient Spatial Representation for Entity-Typing**
Anand Dhoot*, **Aditya Kusupati*** and Soumen Chakrabarti.
Undergraduate Thesis, Computer Science and Engineering, IIT Bombay, 2016 - 17.

SOFTWARE

1. **EdgeML: Machine Learning for Resource-constrained Edge Devices**.
Dennis et al., including **Aditya Kusupati**.
Microsoft Research India, 2017.
Stats: ★ >1,200, 📄 >320, 👁 >300,000, 📦 >4,500.

SELECT AWARDS AND HONORS

- **Best Paper Award** at ENLSP workshop @ NeurIPS '23 2023
- **JUWELS Booster Compute Grant** worth 100K A100 GPU hours 2023
- **Best Poster Award** and a Research Grant worth \$25,000 at Citadel Securities PhD Summit 2023
- Google **Level 3: Accelerate** Research Grant worth \$300,000 extendable up to 1M dollars 2022
- Academic Research **GCP Credit Award** worth \$100,000 2021 - 2023
- **Expert Reviewer** for ICML '21 2021
- **Best Paper Runner-Up Award** at BuildSys '19 2019
- Young Researcher at Heidelberg Laureate Forum (**HLF '19**) with Romberg Grant & MSR Travel Grant 2019
- **Facebook AI Research International Scholarship** for DPhil at VGG, Oxford (2019 - 22, declined) 2019
- IIT Bombay CSE **Teaching Assistant of the month** (Feb '16 and Feb '17) award 2016 - 2017
- **All India Rank 44** in JEE Advanced (IIT-JEE) 2013 among 150,000 candidates qualified from 1.5 million 2013
- Gold Medal and rank **6 out of top 40** in India at OCSC for International Chemistry Olympiad '13 2013
- KVPY Fellowship from Government of India - All India Rank 22. 2011
- NTSE Scholarship from Government of India. 2008
- **Best/Top/Outstanding Reviewer** award for NeurIPS '19, '20, '22; ICML '20, '21; CVPR '21 & ICLR '22

TALKS

- **Matryoshka Principles for Adaptive Intelligence**
 - UCLA NLP Seminar Series May 2025
 - Together AI Talk Series May 2025
 - Next Generation Modeling @ Google March 2025
 - Microsoft Research March 2025
- **Towards Adaptive Intelligence**
 - Exa AI September 2024
 - Sage Bionetworks August 2024

– University of Washington	<i>June 2024</i>
– Microsoft Research AI Frontiers	<i>May 2024</i>
– Snowflake	<i>May 2024</i>
– Google DeepMind	<i>May 2024</i>
– UT Austin Computer Science Colloquium	<i>April 2024</i>
– NVIDIA Research	<i>April 2024</i>
– Google Research	<i>April 2024</i>
– Microsoft Research India	<i>March 2024</i>
– IIT Bombay Computer Science & C-MInDS	<i>March 2024</i>
– Harvard University Computer Science & Kempner Institute Lecture	<i>February 2024</i>
– Columbia University Computer Science Colloquium	<i>February 2024</i>
• Indexing the World	
– Hazy Research Lab @ Stanford	<i>November 2023</i>
– Scaled Foundations	<i>October 2023</i>
– MIT Vision and Graphics Seminar	<i>March 2023</i>
– Harvard Machine Learning Foundations Seminar	<i>March 2023</i>
– Google Research India	<i>February 2023</i>
– H2Lab Seminar @ UW CSE	<i>January 2023</i>
• Matryoshka Representation Learning	
– Jina AI	<i>March 2024</i>
– ThursdAI	<i>February 2024</i>
– Weaviate Podcast	<i>February 2024</i>
– Mosaic ML	<i>June 2023</i>
– Neural Information Processing Systems (NeurIPS)	<i>December 2022</i>
– Pinterest Labs	<i>September 2022</i>
– Perception Spotlight @ Google Research	<i>August 2022</i>
– DeepPhenomena @ Google Research	<i>August 2022</i>
– Image Understanding @ Google Research	<i>June 2022</i>
• LLC: Accurate, Multi-purpose Learnt Low-dimensional Binary Codes	
– Image Understanding @ Google Research	<i>February 2022</i>
– Neural Information Processing Systems (NeurIPS)	<i>December 2021</i>
– Microsoft Research India	<i>November 2021</i>
– UC Berkeley Computer Vision Seminar	<i>November 2021</i>
– University of Washington CSE Colloquium	<i>October 2021</i>
• Soft Threshold Weight Reparameterization for Learnable Sparsity	
– International Conference on Machine Learning (ICML)	<i>July 2020</i>
– NVIDIA Research	<i>July 2020</i>
– Deep Learning: Classics and Trends	<i>June 2020</i>
• The Edge of Machine Learning	
– University of Washington CSE Colloquium & Sensor Systems Seminar	<i>October 2019</i>
– VGG @ Oxford University, UK	<i>April 2019</i>
– Microsoft Research Redmond	<i>March 2019</i>
– Microsoft Research India	<i>August 2018</i>
• The Extremes of Machine Learning	
– Microsoft Bing	<i>March 2019</i>

TEACHING EXPERIENCE

• Co-instructor – Computer Science and Engineering, University of Washington	
– CSE 493G1/599G1: Deep Learning w/ Prof. Ali Farhadi	<i>Fall 2023</i>

- CSE 493G1/599G1: Deep Learning w/ Prof. Ranjay Krishna Spring 2023
- *Undergraduate Teaching Assistantship* – Computer Science and Engineering, IIT Bombay
 - Digital Logic Design - Prof. Supratik Chakraborty - **TA of the month, Feb '17** Spring 2017
 - Software Systems Lab - Prof. Sharat Chandran Autumn 2016
 - Digital Logic Design - Prof. Supratik Chakraborty - **TA of the month, Feb '16** Spring 2016
 - Computer Programming and Utilisation - Prof. Varsha Apte Autumn 2015
 - Computer Programming and Utilisation - Prof. Kavi Arya Spring 2015

PROFESSIONAL SERVICE

- *Area Chair*: NeurIPS 2025, COLM 2025.
- *Reviewing*: IEEE TPAMI, TMLR, NeurIPS (2019 - present), ICML (2020 - present), ICLR (2021 - present), CVPR (2021 - present), ICCV/ECCV (2021 - present), AAAI 2025.
- *Workshop Organization*
 - ML in India Social NeurIPS 2021
 - Rethinking ML Papers ICLR 2021
- *Mentorship*
 - Students (Position → Next Placement)
 - * Sanjay Subramanian
Student Researcher, Google DeepMind 2025 -
 - * Sahil Goyal
Pre-doc Researcher, Google DeepMind 2024 -
 - * Pranav Nair [C.24]
Pre-doc Researcher, Google DeepMind → PhD Student @ Harvard CS 2024 -
 - * Jeevesh Juneja
Pre-doc Researcher, Google DeepMind 2024 -
 - * Puranjay Datta [C.24]
Pre-doc Researcher, Google DeepMind 2024 -
 - * Gagan Jain [C.20]
Pre-doc Researcher, Google DeepMind 2023 -
 - * Ethan Shen [C.22, W.2]
BS Student, UW CSE → PhD Student, UW CSE 2023 - 25
 - * Devvrit [C.19, P.2]
PhD Student, UT Austin CS 2023 - 25
 - * Alan Fan [C.22, C.12, C.15]
BS Student, UW CSE → Software Engineer @ LinkedIn 2023 - 24
 - * Pruthvi Raju
Software Engineer, Google 2022 - 23
 - * Sharan Ranjit [C.12]
MS student, UW ECE → Machine Learning Engineer @ Autodesk 2022 - 23
 - * Venkata Sailesh Sanampudi
Software Engineer, Google 2022 - 24
 - * Umangi Jain [C.17]
Pre-doc Researcher, Google Research → PhD Student @ UofT CS 2022 - 23
 - * Avishree Khare
Software Engineer, Google → Research Fellow @ Microsoft Research 2022
 - * Gantavya Bhatt [C.9]
PhD Student, UW ECE 2022 - 23
 - * Aniket Rege [C.9, C.12, C.25]
MS Thesis, UW ECE → PhD Student @ UW–Madison CS 2022 - 23
 - * William Howard-Snyder [C.9]
BS/MS Student, UW CSE Fall 2021
 - * Oindrila Saha [C.5]
Research Fellow, Microsoft Research → PhD Student @ UMass CS 2019 - 21
 - * Sachin Goyal
Research Fellow, Microsoft Research → PhD Student @ CMU MLD 2019 - 21

- * [Nilesh Gupta](#) [C.3, C.18, P.2]
Research Fellow, Microsoft Research → PhD Student @ UT Austin CS *2019 - 20*
- * [Sahil Bhatia](#)
Research Fellow, Microsoft Research → PhD Student @ UC Berkeley EECS *2018 - 20*
- * [Sheshansh Agrawal](#)
Bachelor's Thesis, IIT Bombay → RSDE @ Microsoft Research *2018 - 19*
- * [Manish Singh](#) [C.1]
Bachelor's Thesis, IIT Delhi → PhD Student @ MIT EECS *2017 - 18*
- 🏆 Best Undergraduate Thesis Award (2018), IIT Delhi**
- New In ML session @ NeurIPS '19 *2019*
- MSR India Summer Workshop 2018: Machine Learning on Constrained Devices *Summer 2018*
- *Faculty Recruiting Liaison* - Paul G. Allen School of CSE, University of Washington *2020 - 2022*
- *Student Area Chair (ML/AI): PhD Admissions* - Paul G. Allen School of CSE, University of Washington *2020 - 2022*
- *Co-Founder & Organizing Committee Member* - Allen School PhD Pre-Application Mentorship Service (PAMS) *2021*
- *Co-Founder & Co-Lead* - Allen School PhD Pre-Application Review Service (PARS) *2020*
- *Department General Secretary* - Computer Science and Engineering, IIT Bombay *2016 - 2017*